

Microelectronics

沈威宇

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1 Microelectronics

I Signal

i Signal

A signal is a function that conveys information about a phenomenon. Signal processing is usually most conveniently performed by electronic systems on voltage or current signals, especially the former. Only voltage and current signals are studied here.

ii Signal source

A signal source is typically represented in the Thévenin form, a voltage source in a series connection with a resistance, or the Norton form, a current source in a parallel connection with a resistance.

iii Cascade

Output of a component as the input of another.

iv Transducer

A transducer is a device that convert a signal to a voltage or current signal or vice versa.

v Signal processing

Signal processing includes analog signal processing (ASP), analog-to-digital converter (ADC), digital signal processor (DSP), digital-to-analog converter (DAC), amplifier, etc.

vi Coupling capacitor

A capacitor connected in series between stages to pass AC signal while blocking DC.

vii Symbol convention

- Lowercase character with uppercase subscript means total signal, e.g. $i_C(t)$.
- Lowercase character with lowercase subscript means AC signal, e.g. $i_c(t)$.
- Uppercase character with uppercase subscript means DC signal, e.g. I_C .
- Uppercase character with lowercase subscript means peak signal, e.g. I_c .

Their relation is

$$i_C(t) = i_c(t) + I_C$$

$$i_c(t) = I_c \sin(\omega t + \varphi)$$

viii differential and common-mode signal

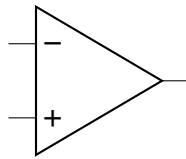
For a component with two input signals, the differential signal (AC signal) v_d is the difference between the two input signals, and the common-mode signal (DC signal) v_{cm} is the algebraic average of the two input signals.

II Amplifier (amp)

i Amplifier

An electronic device which amplifies an electrical signal by a scale called amplifier gain A and optionally convert voltage signal to current signal or vice versa.

ii Diagram



iii Two-port model

An amplifier can be modeled as a two port model, where an one input port and one output port is related by the amplifier, which acts as a controlled source controlled by the input port voltage or current which enforces voltage or current on the output port.

iv Circuit model of voltage amplifier

A voltage amplifier draws current from a source v_s with internal resistance R_s with input resistance R_i and voltage v_{in} across it and supplies $v_c = \mu v_{in}$ to a load network with output resistance R_o and load R_L , where μ , also denoted A_{vo} , is open-loop voltage gain factor.

$$\frac{v_{in}}{v_s} = \frac{R_i}{R_s + R_i}$$

$$\frac{v_c}{v_{in}} = \mu$$

$$\frac{v_{out}}{v_c} = \frac{R_L}{R_o + R_L}$$

Voltage gain:

$$A_v = \frac{v_{out}}{v_{in}} = \mu \frac{R_i R_L}{(R_s + R_i)(R_o + R_L)}$$

$|A| < |\mu|$ due to loading effects.

A good amplifier requires small loading effect:

$$R_i \gg R_s, \quad R_o \ll R_L, \quad A_v \approx \mu$$

v Circuit model of current amplifier

A current amplifier draws current from a current source i_s with internal resistance R_s in parallel with input resistance R_i and current i_{in} through it and supplies $i_c = \beta i_{in}$ by CCCS to a load network with output resistance R_o in parallel and load R_L , where β , also denoted A_{cs} , is short-circuit current gain factor.

$$\begin{aligned}\frac{i_{in}}{i_s} &= \frac{R_s}{R_s + R_i} \\ \frac{i_c}{i_{in}} &= \beta \\ \frac{i_{out}}{i_c} &= \frac{R_o}{R_o + R_L}\end{aligned}$$

Current gain:

$$A_c = \frac{i_{out}}{i_{in}} = \beta \frac{R_s R_o}{(R_s + R_i)(R_o + R_L)}$$

$|A| < |\beta|$ due to loading effects.

A good amplifier requires small loading effect:

$$R_i \ll R_s, \quad R_o \gg R_L, \quad A_c \approx \beta$$

vi Circuit model of transconductance amplifier

A transconductance amplifier draws current from a voltage source v_s with internal resistance R_s in series with input resistance R_i and current i_{in} through it and supplies $i_c = g v_{in}$ by VCCS to a load network with output resistance R_o in parallel and load R_L , where μ is open-loop transconductance factor.

$$\begin{aligned}\frac{v_{in}}{v_s} &= \frac{R_i}{R_s + R_i} \\ \frac{i_c}{v_{in}} &= g \\ \frac{i_{out}}{i_c} &= \frac{R_o}{R_o + R_L}\end{aligned}$$

Gain:

$$A_g = \frac{i_{out}}{v_{in}} = g \frac{R_i R_o}{(R_s + R_i)(R_o + R_L)}$$

$|A| < |g|$ due to loading effects.

A good amplifier requires small loading effect:

$$R_i \gg R_s, \quad R_o \gg R_L, \quad A_g \approx g$$

vii Circuit model of transresistance amplifier

A transresistance amplifier draws current from a current source i_s with internal resistance R_s in parallel with input resistance R_i and current i_{in} through it and supplies $v_c = r i_{in}$ by CCVS to a load network with output resistance R_o in series and load R_L , where r is short-circuit transresistance factor.

$$\frac{i_{in}}{i_s} = \frac{R_s}{R_s + R_i}$$

$$\frac{v_c}{i_{in}} = r$$

$$\frac{v_{out}}{v_c} = \frac{R_L}{R_o + R_L}$$

Gain:

$$A_r = \frac{v_{out}}{i_{in}} = \beta \frac{R_s R_L}{(R_s + R_i)(R_o + R_L)}$$

$|A| < |r|$ due to loading effects.

A good amplifier requires small loading effect:

$$R_i \ll R_s, \quad R_o \ll R_L, \quad A_c \approx r$$

viii Amplifier gain in decibels (dB)

A power gain A_p in dB is defined as

$$G_p = 10 \log_{10}(A_p)$$

A voltage gain A_v in dB is defined as

$$G_v = 20 \log_{10}(A_v)$$

A current gain A_c in dB is defined as

$$G_c = 20 \log_{10}(A_c)$$

ix Saturation

The linear characteristic only remains within a range. A positive and negative saturation levels can be observed.

x S parameter

If the energy enters through port i and emerges from port j , the S parameter is named S_{ji} .

xi Unilateral amplifier

An unilateral amplifier is an amplifier in which the signal transfer is effectively one-way, that is, the output signal is only dependent on the input signal but not what is connected to the output terminal. That is for each output port j and each input port i , $S_{ij} = 0$.

xii Power supply

To amplify signal, power supplies are needed, but often omitted in the diagram.

- V_{CC+} or V_{CC} : positive supply voltage, denoted on the + side of an op-amp or omitted;
- V_{CC-} or $-V_{EE}$: negative supply voltage, denoted on the – side of an op-amp or omitted.

The DC power delivered is the sum of power from the positive and negative power supply terminals V_{CC} and V_{EE}

$$P_{DC} = V_{CC}I_{CC} + V_{EE}I_{EE}$$

Suppose input and output signals have power P_I and P_L .

$$P_{DC} + P_I = P_L + P_{\text{dissipated}}$$

where $P_{\text{dissipated}}$ is the power dissipated.

xiii Classification based on frequency response

A band-pass amplifier is also called tuned amplifier. A low-pass amplifier is also called DC amplifier. A amplifier directly coupled with a coupling capacitor, called capacitively coupled, is called AC amplifier.

III Operational Amplifier (op amp or op-amp)

i Definition

An op amp has five terminals:

- v_+ : non-inverting input voltage of the terminal at the + sign;
- v_- : inverting input voltage of the terminal at the – sign;
- v_o : output voltage of the terminal at the opposite side of inputs;
- V_{CC+} or V_{CC} : positive supply voltage, denoted on the + side of an op-amp or omitted;
- V_{CC-} or $-V_{EE}$: negative supply voltage, denoted on the – side of an op-amp or omitted.

$$v_o = A(v_+ - v_- - v_{os})$$

where A is open-loop gain or open-circuit gain, and v_{os} is input offset voltage.

$$v_{ld} = v_+ - v_-$$

$$v_{lcm} = \frac{v_+ + v_-}{2}$$

ii Ideal op amp

- Infinite input impedance: No signal current flowing into v_+ and v_- .
- Infinite open-loop gain
- Infinite bandwidth: open-loop gain remains constant in all frequencies.
- Zero output impedance: The amplifier is unilateral.
- Zero input offset voltage: $v_{os} = 0$, which implies zero common-mode gain, i.e., infinite common-mode rejection.

This is a linear element with inputs v_+ and v_- and output v_o .

We can construct a virtual short circuit between v_+ and v_- for ideal op amp.

Note that while $v_+ - v_-$, v_o may be any voltage since $A = \infty$ and $0 * \infty$ can be any number in $[0, \infty]$ in limit sense.

iii Feedback configuration

To have stable signal, op amp in a circuit is connected with a negative feedback configuration, which consists of the op amp and a connection from v_o to either of the input terminals with a resistance on it, called feedback connection. When the feedback connection connects to v_- , it is called inverting configuration. When the feedback connection connects to v_+ , it is called non-inverting configuration.

iv Inverting amplifier

An inverting amplifier consists of an op amp where input voltage v_i is connected to v_- through an input resistance R_i (current from v_i to v_-), v_+ is connected to ground, and v_o is fed back to v_- through a feedback resistance R_f (current from v_- to v_o).

$$\begin{aligned} v_+ &= v_- = 0 \\ i_i &= \frac{v_i}{R_i} = i_f = -\frac{v_o}{R_f} \\ v_o &= -\frac{R_f}{R_i} v_i \end{aligned}$$

The closed-loop gain G is defined as

$$G = \frac{v_o}{v_i} = -\frac{R_f}{R_i}$$

When $-G < 1$, it's called an attenuator.

Effect of finite open-loop gain:

$$\begin{aligned} v_+ &= 0 \\ v_o &= -A v_- \\ i_i &= \frac{v_i - v_-}{R_i} = \frac{v_i + \frac{v_o}{A}}{R_i} = i_f = -\frac{v_o + \frac{v_o}{A}}{R_f} \\ G &= \frac{v_o}{v_i} = -\frac{R_f}{R_i \left(1 + \frac{1 + \frac{R_f}{R_i}}{A} \right)} = \frac{G_{\text{ideal}}}{1 + \frac{1 - G_{\text{ideal}}}{A}} \\ \lim_{A \rightarrow \infty} G &= -\frac{R_f}{R_i} = G_{\text{ideal}} \end{aligned}$$

v Transresistance amplifier

A transresistance amplifier consists of an op amp where input current i_i is connected to v_- , v_+ is connected to ground, and v_o is fed back to v_- through two resistances R_o (current from x to v_o) and R_f (current from v_- to x) in series connection, with the node x between R_o and R_f connected to ground through a resistance R_g (current from x to ground).

$$v_x = -R_f i_i$$

$$i_g = -\frac{R_f}{R_g} i_i$$

$$i_o = \left(1 + \frac{R_f}{R_g}\right) i_i$$

$$v_o = -R_f i_i - R_o i_o = -\left(R_f + R_o + \frac{R_f R_o}{R_g}\right) i_i$$

$$\frac{v_o}{i_i} = -\left(R_f + R_o + \frac{R_f R_o}{R_g}\right)$$

vi Weighted summing amplifier

A weighted summing amplifier consists of an op amp, input voltages $v_1, v_2, \dots, v_n$, n input resistors $R_1, R_2, \dots, R_n$, and a feedback resistor R_f , where for any $k \in \mathbb{N} \wedge k \leq n$, v_k is connected to v_- through R_k , v_+ is connected to ground, and v_o is fed back to v_- through R_f .

$$i_k = \frac{v_k}{R_k}$$

$$\sum_{k=1}^n i_k = i_f = \frac{-v_o}{R_f}$$

$$v_o = -R_f \sum_{k=1}^n \frac{v_k}{R_k}$$

When $R_k = R$ for all $k \in \mathbb{N} \wedge k \leq n$:

$$v_o = -\frac{R_f}{R} \sum_{k=1}^n v_k$$

vii Non-inverting amplifier

A non-inverting (or noninverting) amplifier consists of an op amp where input voltage v_i is connected to v_+ , v_- is connected ground through a ground resistance R_g (current from v_- to ground), and v_o is fed back to v_- through a feedback resistance R_f (current from v_o to v_-).

$$v_+ = v_- = v_i$$

$$i_g = \frac{v_i}{R_g} = i_f = \frac{v_o - v_i}{R_f}$$

$$v_o = \left(1 + \frac{R_f}{R_g}\right) v_i$$

The closed-loop gain G is defined as

$$G = \frac{v_o}{v_i} = 1 + \frac{R_f}{R_g}$$

Effect of finite open-loop gain:

$$\begin{aligned} v_+ &= v_i \\ v_o &= A(v_i - v_-) \\ v_- &= v_i - \frac{v_o}{A} \\ i_g = \frac{v_-}{R_g} &= \frac{v_i - \frac{v_o}{A}}{R_g} = i_f = \frac{v_o - v_-}{R_f} = \frac{v_o - v_i + \frac{v_o}{A}}{R_f} \\ G = \frac{v_o}{v_i} &= \frac{1 + \frac{R_f}{R_g}}{1 + \frac{R_f}{A}} = \frac{G_{\text{ideal}}}{1 + \frac{G_{\text{ideal}}}{A}} \\ \lim_{A \rightarrow \infty} G &= 1 + \frac{R_f}{R_g} = G_{\text{ideal}} \end{aligned}$$

viii Unity gain amplifier, unity gain buffer, buffer amplifier, or voltage follower

A unity gain amplifier consist of an op amp where input voltage v_i is connected to v_+ , and v_o is fed back to v_- .

$$v_o = v_- = v_+ = v_i$$

ix Difference amplifier or differential amplifier

A difference amplifier consists of an op amp where first input voltage v_1 is connected to v_- through an input resistance R_1 (current from v_1 to v_-), second input voltage v_2 is connected to v_+ through an input resistance R_2 (current from v_2 to v_+), v_+ is connected to ground through a ground resistance R_g (current from v_+ to ground), and v_o is fed back to v_- through a feedback resistance R_f (current from v_- to v_o). Denote common-mode signal $v_{cm} = \frac{v_1 + v_2}{2}$ and differential signal $v_d = v_2 - v_1$.

$$\begin{aligned} v_o &= -\frac{R_f}{R_1}v_1 + \frac{R_g}{R_2 + R_g} \left(1 + \frac{R_f}{R_1} \right) v_2 \\ i_1 &= \frac{1}{R_1} \left(v_1 - \frac{R_g}{R_2 + R_g} v_2 \right) \\ i_2 &= \frac{v_2}{R_2 + R_g} \end{aligned}$$

Common-mode gain A_{cm} and differential gain A_d , defined by

$$v_o = v_{cm}A_{cm} + v_dA_d,$$

are

$$A_{cm} = \frac{R_g}{R_2 + R_g} \left(1 - \frac{R_2}{R_g} \frac{R_f}{R_1} \right)$$

$$A_d = \frac{R_g}{R_2 + R_g} \left(1 + \frac{R_2 + 2R_g}{R_g} \frac{R_f}{R_1} \right)$$

Common-mode rejection ratio (CMRR) is defined as

$$\text{CMRR} = 20 \log \left| \frac{A_d}{A_{cm}} \right|$$

To make $A_{cm} = 0$, we need

$$\frac{R_f}{R_1} = \frac{R_g}{R_2},$$

at which

$$A_d = \frac{R_f}{R_1} = \frac{R_g}{R_2}$$

Differential input resistance R_d , defined as, when we connect v_1 and v_2 to a source v_d ($v_d = v_2 - v_1$ and $i_d = i_2 = -i_1$),

$$\frac{v_d}{i_d},$$

is

$$R_d = R_1 + R_2$$

Thus, high differential gain hurts differential input resistance, which lower the voltage applied to the amplifier since source has internal resistance.

Proof. Use superposition. With v_1 enabled and v_2 disabled,

$$v_+ = v_- = 0$$

$$i_2 = i_g = 0$$

$$i_1 = \frac{v_1}{R_1} = i_f$$

$$v_o = -i_f R_f = -\frac{R_f}{R_1} v_1$$

With v_2 enabled and v_1 disabled,

$$i_2 = i_g = \frac{v_2}{R_2 + R_g}$$

$$v_+ = v_- = i_g R_g = \frac{R_g}{R_2 + R_g} v_2$$

$$i_1 = -\frac{v_-}{R_1} = -\frac{R_g}{R_1(R_2 + R_g)} v_2 = i_f$$

$$v_o = v_- - i_f R_f = \frac{R_g}{R_2 + R_g} v_2 + \frac{R_g R_f}{R_1(R_2 + R_g)} v_2 = \frac{R_g}{R_2 + R_g} \left(1 + \frac{R_f}{R_1} \right) v_2$$

Sum:

$$v_o = -\frac{R_f}{R_1} v_1 + \frac{R_g}{R_2 + R_g} \left(1 + \frac{R_f}{R_1} \right) v_2$$

$$i_1 = \frac{v_1}{R_1} - \frac{R_g}{R_1(R_2 + R_g)} v_2 = \frac{1}{R_1} \left(v_1 - \frac{R_g}{R_2 + R_g} v_2 \right)$$

$$i_2 = \frac{v_2}{R_2 + R_g}$$

$$v_1 = v_{cm} - \frac{v_d}{2}$$

$$v_2 = v_{cm} + \frac{v_d}{2}$$

$$\begin{aligned} v_o &= -\frac{R_f}{R_1} \left(v_{cm} - \frac{v_d}{2} \right) + \frac{R_g}{R_2 + R_g} \left(1 + \frac{R_f}{R_1} \right) \left(v_{cm} + \frac{v_d}{2} \right) \\ &= \frac{R_g}{R_2 + R_g} \left(1 - \left(\frac{R_2 + R_g}{R_g} - 1 \right) \frac{R_f}{R_1} \right) v_{cm} + \frac{R_g}{R_2 + R_g} \left(1 + \left(\frac{R_2 + R_g}{R_g} + 1 \right) \frac{R_f}{R_1} \right) v_d \\ &= \frac{R_g}{R_2 + R_g} \left(1 - \frac{R_2 R_f}{R_g R_1} \right) v_{cm} + \frac{R_g}{R_2 + R_g} \left(1 + \frac{R_2 + 2R_g}{R_g} \frac{R_f}{R_1} \right) v_d \end{aligned}$$

$$A_{cm} = \frac{R_g}{R_2 + R_g} \left(1 - \frac{R_2 R_f}{R_g R_1} \right)$$

$$A_d = \frac{R_g}{R_2 + R_g} \left(1 + \frac{R_2 + 2R_g}{R_g} \frac{R_f}{R_1} \right)$$

To make $A_{cm} = 0$, we need

$$1 = \frac{R_2 R_f}{R_g R_1}$$

$$\frac{R_f}{R_1} = \frac{R_g}{R_2}$$

at which

$$\begin{aligned} A_d &= \frac{R_g}{R_2 + R_g} \left(1 + \frac{R_2 + 2R_g}{R_g} \frac{R_f}{R_1} \right) \\ &= \frac{R_g}{R_2 + R_g} \left(1 + \frac{R_2 + 2R_g}{R_2} \right) \\ &= \frac{R_g}{R_2} \end{aligned}$$

Differential input resistance:

$$i_d = \frac{v_+}{R_g} = \frac{v_2 - v_+}{R_2} = \frac{v_+ - v_2 + v_d}{R_1}$$

$$R_2 v_+ = R_g v_2 - R_g v_+$$

$$R_1 v_2 - R_1 v_+ = R_2 v_+ - R_2 v_2 + R_2 v_d$$

$$v_2 = \left(\frac{R_2 + R_g}{R_g} \right) v_+$$

$$(R_1 + R_2)(v_2 - v_+) = R_2 v_d$$

$$v_+ = \frac{R_g}{R_1 + R_2} v_d$$

$$i_d = \frac{1}{R_1 + R_2} v_d$$

$$R_d = \frac{v_d}{i_d} = R_1 + R_2$$

□

x Instrumentation amplifier

An instrumentation amplifier consists of a difference amplifier (second stage) with outputs of two identical non-inverting amplifiers (first stage) as inputs, which behaves the same as the difference amplifier but with both common-mode and differential gains multiplied by the gain of the first stage. It can achieve higher differential gain than difference amplifier under same differential input resistance. However, common-mode signal is also amplified in the first stage, so CMRR of second stage must be good, i.e., the resistances must be precise.

xi Inverting configuration with general impedance

Consider an op amp where input voltage v_i is connected to v_- through an input impedance Z_i (current from v_i to v_-), v_+ is connected to ground, and v_o is fed back to v_- through a feedback impedance Z_f (current from v_- to v_o).

Laplace domain:

$$\begin{aligned}V_+ &= V_- = 0 \\I_i &= \frac{V_i}{Z_i} = I_f = -\frac{V_o}{Z_f} \\V_o &= -\frac{Z_f}{Z_i} V_i\end{aligned}$$

xii Inverting integrator or Miller integrator non-feasible configuration

An inverting integrator non-feasible configuration consists of an op amp where input voltage v_i is connected to v_- through an input resistance R , v_+ is connected to ground, and v_o is fed back to v_- through a feedback capacitance C .

$v_o(0^-)$ is the initial voltage across C (v_o relative to v_-).

Time domain:

$$\begin{aligned}v_+ &= v_- = 0 \\i_i &= \frac{v_i}{R} = i_f = -C \frac{dv_o}{dt} \\v_o &= v_o(0^-) - \frac{1}{RC} \int_0^t v_i(\tau) d\tau\end{aligned}$$

Laplace domain:

$$\begin{aligned}V_+ &= V_- = 0 \\I_i &= \frac{V_i}{R} = I_f = -CsV_o + Cv_o(0^-) \\V_o &= \frac{1}{s}v_o(0^-) - \frac{1}{RCs}V_i\end{aligned}$$

Transfer function: Infinite DC gain. Thus, any tiny DC component in the input signal will blow this out.

$$H(s) = -\frac{1}{RCs}$$

xiii Inverting integrator or Miller integrator feasible configuration

An inverting integrator feasible configuration consists of an op amp where input voltage v_i is connected to v_- through an input resistance R , v_+ is connected to ground, and v_o is fed back to v_- through a feedback capacitance C and feedback resistance R_f in parallel connection.

$v_o(0^-)$ is the initial voltage across C (v_o relative to v_-).

Laplace domain:

$$V_+ = V_- = 0$$
$$I_i = \frac{V_i}{R} = I_f = - \left(\frac{1}{R_f} + Cs \right) V_o + C v_o(0^-) = - \frac{1 + R_f Cs}{R_f} V_o + C v_o(0^-)$$
$$V_o = \frac{R_f C}{1 + R_f Cs} v_o(0^-) - \frac{\frac{R_f}{R}}{1 + R_f Cs} V_i$$

Transfer function: Butterworth first-order low-pass with cutoff frequency $\frac{1}{R_f C}$ and DC gain $\frac{R_f}{R}$.

$$H(s) = - \frac{\frac{R_f}{R}}{1 + R_f Cs}$$

xiv inverting differentiator

An inverting differentiator consists of an op amp where input voltage v_i is connected to v_- through an input capacitor C_i , v_+ is connected to ground, and v_o is fed back to v_- through a feedback resistor R_f .

Time domain:

$$i_i = C_i \frac{dv_i}{dt} = i_f = \frac{-v_o}{R_f}$$
$$v_o = R_f C_i \left(v_i(0^-) - \int_0^t v_i(\tau) d\tau \right)$$

Laplace domain:

$$i_i = C_i s v_i - C_i v_i(0^-) = i_f = \frac{-v_o}{R_f}$$
$$v_o = R_f C_i (v_i(0^-) - s v_i)$$

xv Active filter

Given a (passive or active) filter v_f of an input v_s and an inverting amplifier, non-inverting amplifier, or unit gain buffer enforcing $v_o = K v_i$, by connecting the ground and v_i of the amplifier to the two terminals of the v_f , the two terminals ground and v_o become an active filter of v_s enforcing $v_o = K v_f$.